# Machine Learning–Based Classification of Spine Health for Predicting Lower Back Pain Conditions

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**Abstract**

Lower back pain is a prevalent condition with a variety of causes, ranging from nerve irritation to muscle strain and joint damage. This project aims to predict whether a person's lower back condition is "Normal" or "Abnormal" using a dataset of 310 observations, each containing 12 numeric features related to spinal attributes. These features include measurements like pelvic tilt, lumbar lordosis angle, and sacral slope. By applying machine learning classification techniques, we aim to analyze these physical attributes and develop a model to classify the back condition accurately. The results of this analysis can potentially aid in the early detection and management of lower back pain, providing healthcare professionals with valuable insights for better diagnosis and treatment planning.

**Keywords**: Lower back pain, machine learning, classification, spinal attributes, prediction.

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# INTRODUCTION

What causes lower back pain? The first factors that come to mind are muscle strains, nerve irritation, or issues with the bones and joints in the lower spine. More advanced cases may involve disc degeneration or damage to the spinal tendons. The source of the pain can vary greatly from person to person and is often influenced by various factors like posture, lifestyle, or even genetics.

While there are a variety of causes for lower back pain, it remains a complex and subjective issue to diagnose. For example, two individuals experiencing similar symptoms may have different underlying causes, and the severity of the pain can range from mild discomfort to debilitating chronic pain. The challenge, therefore, lies in identifying abnormal spinal conditions early, which could potentially prevent further complications. Early diagnosis is especially important for healthcare practitioners, as it allows them to take timely and effective actions that can reduce the risk of chronic conditions and improve patient outcomes.

On the other hand, there are measurable and objective features of the spine that could help assess the severity of lower back pain. Attributes such as pelvic tilt, lumbar lordosis angle, and sacral slope are among the key physical attributes that have been shown to influence the health of the lumbar spine. By analyzing these features, we may be able to classify whether a person's condition is "Normal" or "Abnormal" and thus predict the likelihood of back pain.

For this project, we use a dataset containing physical spine data of **310 observations**, each with **6 numeric attributes** and a **binary class attribute** (Normal/Abnormal). The dataset includes various attributes such as:

- **Pelvic Incidence**: A numeric attribute representing the angle of the pelvis.
- **Pelvic Tilt**: Measures the tilt of the pelvis relative to the body.
- **Lumbar Lordosis Angle**: The angle between the lumbar spine and the horizontal plane.
- **Sacral Slope**: The angle of the sacrum relative to the horizontal plane.
- **Pelvic Radius**: Represents the radius of the pelvis.
- **Degree of Spondylolisthesis**: Measures the degree of vertebra displacement in the spine.

The **Class Attribute** is binary: "Normal" or "Abnormal," indicating whether the individual's lower back condition is considered normal or shows abnormal symptoms.

The goal of our analysis is to predict whether an individual's lower back condition is "Normal" or "Abnormal" based on these physical attributes using machine learning classification techniques. By developing a reliable predictive model, we aim to assist healthcare professionals in making informed decisions and providing better patient care.

The machine learning models are implemented using **KNIME**, and the workflow is organized into several key stages:

1. **Data Exploration**: This step includes examining the dataset and understanding

the distribution of key features, especially the class labels. Missing values are also identified and handled accordingly.

2. **Data Preprocessing and Feature Selection**: Various preprocessing tasks, such as missing value imputation, normalization, and column filtering, are applied to make the dataset suitable for modeling. Linear correlation and other statistical techniques are used to select relevant features for model training, ensuring that unnecessary or redundant features are removed.
3. **Data Splitting**: The dataset is split into training and test sets (70% vs. 30%) to ensure that the model is evaluated on unseen data.

   **Handling Class Imbalance with SMOTE**: To address the **class imbalance** in the dataset, we applied **SMOTE** (Synthetic Minority Over- sampling Technique) to the **training set**. **SMOTE** generated synthetic samples for the minority class (**Normal**) by interpolating between existing minority class instances. This helped balance the class distribution and allowed the models to learn effectively from both classes. By balancing the dataset using **SMOTE**, we improved the models' ability to predict both **Abnormal** and **Normal** conditions accurately.
4. **Model Training**: Several machine learning models are trained, including **Random Forest**, **Gradient Boosted Trees**, **Logistic Regression**, **Support Vector Machines (K NEAREST NEIGHBOUR)**, **Decision Trees**, and **Naive Bayes**.
5. **Model Evaluation**: The models are evaluated using various metrics, such as **accuracy**, and the best-performing models are identified based on their ability to predict the class attribute ("Normal" vs. "Abnormal"). The results are visualized to compare model performance and to display the distribution of class predictions.

By employing these machine learning techniques, the goal is to predict lower back conditions accurately, providing healthcare professionals with a reliable tool to assist in the **early diagnosis** and **treatment** of lower back pain.

# DATA EXPLORATION

The first step in our analysis is to explore the dataset in order to understand its characteristics and identify any potential issues that need to be addressed before applying machine learning models. This phase includes statistical analysis, visualizations, and handling of missing values.

## SUMMARY STATISTICS

The dataset contains **310 observations** with **12 numeric features** related to spinal attributes and one binary class label. We begin by examining basic statistics for each feature, such as the minimum, maximum, mean, and standard deviation, to understand the distribution of values. For example:

- **Pelvic Incidence** has a range from **26.15** to **129.83**, with a mean of **60.5**.
- **Pelvic Tilt** ranges from **-6.55** to **49.43**, with a mean of **17.5**.
- **Degree of Spondylolisthesis** shows significant variation with values from **-11.06** to **418.54**.

These statistics help in identifying outliers, extreme values, and features that may need further transformation.

## CLASS DISTRIBUTION

The **class_label** is binary, with two possible values: **Normal** and **Abnormal**. The distribution of the target variable shows that there are **210 instances of Abnormal** and **100 instances of Normal**, indicating an imbalance between the classes. This class imbalance could affect model performance, as most models may tend to favor predicting the majority class (Abnormal). To address this, techniques like oversampling the minority class, undersampling the majority class, or using class weights during model training will be considered.

## MISSING VALUES

The **Missing Value Count** shows that there are **no missing values** in any of the columns, ensuring the dataset is complete. This allows us to proceed without the need for imputation or data removal based on missing values.

## FEATURE DISTRIBUTION

We further examine the feature distributions using histograms. Several features, such as **lumbar lordosis angle**, **pelvic tilt**, and **sacral slope**, exhibit a roughly normal distribution, while others, such as **pelvic radius** and **degree of spondylolisthesis**, show skewed distributions.

- **Pelvic Radius** has a skewed distribution, with a large concentration of values on the lower end (70.08 to 125.88) and a few outliers extending to higher values (up to 163.07).
- **Degree of Spondylolisthesis** also shows a skewed distribution, with a few extreme values indicating cases of severe spinal misalignment.

This insight indicates that some features may require transformations (e.g., logarithmic scaling) to make their relationships with the target variable more linear and to reduce the influence of outliers.

## VISUAL INSPECTION

Visual inspection of the dataset through bar charts and histograms allows us to identify any anomalies, skewness, or outliers in the features.

- A **bar chart** shows the distribution of the class labels (**Normal vs. Abnormal**), confirming the class imbalance.
- **Histograms** for numerical features provide further insights into the spread and distribution of values,

indicating potential preprocessing steps for skewed features.

## DATA PREPROCESSING AND FEATURE SELECTION

The **Data Pre-Processing** phase is crucial for preparing the dataset for machine learning. During this phase, we carried out multiple steps to clean, transform, and optimize the dataset, ensuring that it is in the best possible form for model training. Below is a detailed explanation of the key steps applied to the **Lower Back Pain Symptoms Dataset**.

## HANDLING MISSING VALUES

Missing values can introduce bias or errors into the machine learning model, so it is essential to address them before proceeding with the analysis.

- **Numerical Features**: The **Missing Value** node was used to handle missing values in numerical columns. For the numeric columns, the missing values were replaced by the **mean** of the respective column. This is a standard method of imputation for numerical data and ensures that the mean of each feature is maintained, preventing any data loss.
- **String Features**: For categorical (string) features, we opted to apply **"Do nothing"** for missing values. As per our dataset, there were no critical missing values in categorical

columns that would necessitate imputation, so this step was left as default Handling missing values ensures that we do not lose valuable data and that the dataset remains intact, with no rows removed.

## NORMALIZATION

Machine learning algorithms, especially distance-based models like **k-NN** or **K NEAREST NEIGHBOUR**, perform better when all features are on the same scale. Therefore, we applied **Normalization** to ensure all numeric features fall within the same range.

- **Normalization Method**: The **Min-Max Normalization** method was applied to scale all numeric features between 0 and 1. This method preserves the relationships in the data while adjusting for differences in scale.

Features Normalized:

- Pelvic Incidence
- Pelvic Tilt
- Lumbar Lordosis Angle
- Sacral Slope
- Pelvic Radius
- Degree of Spondylolisthesis

The **Min-Max** scaling ensures that all features contribute equally to the model, regardless of their original scale.

## FEATURE SELECTION

In machine learning, not all features contribute equally to making predictions. Irrelevant or redundant features can negatively affect the performance of the model by increasing complexity or causing overfitting. To enhance model performance, we

applied **Feature Selection** to retain only the most relevant features.

## LINEAR CORRELATION

The first step in feature selection was to calculate the correlation between each pair of features using the **Linear Correlation** node. This analysis helped identify pairs of features that were highly correlated, meaning

their correlation coefficient was close to 1 or -1. For example, **Pelvic Incidence** and **Pelvic Tilt** were found to be highly correlated, which could lead to redundancy in the model. To address this, we decided to exclude one of them.

## CORRELATION FILTER

To further reduce redundancy, we applied the **Correlation Filter** node with a correlation threshold of
**0.7**. This ensured that any features with a correlation higher than 0.7 were excluded. As a result, we removed the following features:

- **Lumbar Lordosis Angle**: This feature was highly correlated with **Pelvic Incidence** and **Pelvic Tilt**, so we excluded it to avoid multicollinearity.
- **Sacral Slope**: Similar to **Lumbar Lordosis Angle**, this feature was also excluded to reduce redundancy and improve model efficiency.

By removing these highly correlated features, we improved the interpretability of the model and reduced the chances of overfitting, ensuring that only the most independent and relevant features were used for prediction.

## COLUMN FILTERING

The **Column Filter** node was used to further refine the dataset by excluding features that are either irrelevant or redundant. After applying feature selection and correlation filtering, we removed the following columns:

Removed Columns:

- Lumbar Lordosis Angle: As explained earlier, it was highly correlated with other features and thus redundant.
- Sacral Slope: Removed to avoid multicollinearity.

Remaining Columns:

After feature selection and column filtering, the final dataset includes the following relevant features:

- Pelvic Incidence
- Pelvic Tilt
- Pelvic Radius
- Degree of Spondylolisthesis

These remaining features are deemed the most relevant for predicting the class label ("Normal" or "Abnormal"), and they are retained for the model training phase.

The dataset now contains **310 observations** and **9 features**, and it is ready for training machine learning models. This clean dataset ensures that our models are trained on relevant, independent, and well-scaled features, providing the best chance for accurate predictions.

# DATA SPLITTING

In machine learning, it's essential to split the dataset into two subsets: one for training the model and the other for testing its performance. This allows us to evaluate the model's generalizability and how well it performs on unseen data.

For this project, we used the **Partitioning** node in KNIME to split the dataset into training and testing sets. The dataset was divided as follows:

- **Training Set**: 70% of the data was used for training the model.
- **Testing Set**: The remaining 30% of the data was set aside for testing the model's performance.

By using this **70% vs. 30%** split, we ensure that the model is trained on a large portion of the data while still being evaluated on a separate, unseen subset. This helps in assessing how well the model can predict new, unseen data, ensuring that it does not overfit to the training set.

# HANDLING CLASS IMBALANCE WITH SMOTE

Class imbalance is a common issue in machine learning tasks, particularly when dealing with datasets where one class is underrepresented compared to the other. In the context of this project, the dataset shows a significant imbalance between the **Normal** and **Abnormal** classes, with **210 instances of Abnormal** and **100 instances of Normal**. This imbalance can lead to biased model predictions, as most models may tend to favor predicting the majority class (**Abnormal**).

To address this issue and ensure the models have equal opportunities to learn from both classes, we applied the **SMOTE** (Synthetic Minority Over-sampling Technique) method. **SMOTE** is an over-sampling technique that generates synthetic examples for the minority class (**Normal**) by interpolating between existing minority class instances. This approach ensures that the model is exposed to a more balanced dataset and is better equipped to predict **Normal** cases alongside **Abnormal** cases. Performance in detecting the minority class (**Normal**). By balancing the dataset using **SMOTE**, the models are now better equipped to provide accurate predictions for both **Abnormal** and **Normal** conditions.

# MODEL TRAINING

To predict whether a lower back condition is **Normal** or **Abnormal**, various machine learning models were applied to the dataset, leveraging the most relevant features identified during the **Feature Selection** phase. These models were trained using **KNIME** and evaluated for their ability to classify the target variable (class_label). Below are the details of the machine learning models chosen for this analysis:

## STEPS INVOLVED IN HANDLING CLASS IMBALANCE WITH SMOTE:

1. **Data Splitting**: The dataset was first split into training and testing sets, maintaining the imbalance in the original class distribution for the test set, but ensuring a more balanced training set.
2. **SMOTE Application**: **SMOTE** was then applied to the **training set** to generate synthetic samples for the minority class (**Normal**). The **SMOTE** nodes in the KNIME workflow are responsible for creating these synthetic instances, which are based on the nearest neighbors of the minority class examples.
3. **Balanced Dataset**: After **SMOTE** is applied, the training data becomes more balanced, with an equal number of **Normal** and **Abnormal** instances. This ensures that the model can learn effectively from both classes, without being biased towards predicting the majority class.
4. **Model Training**: The models, including **Random Forest**, **Logistic Regression**, **K NEAREST NEIGHBOUR**, **Decision Trees**, and others, were trained on the **balanced training set** generated by **SMOTE**.

This method of addressing class imbalance ensures that the model is not overly biased toward the majority class (**Abnormal**) and improves the overall model.

## RANDOM FOREST LEARNER

The **Random Forest** algorithm is an ensemble method that uses multiple decision trees to make predictions. Each tree is trained on a different subset of the data, and the final prediction is determined by combining the results from all trees. The strength of this model lies in its ability to handle large datasets with complex relationships and its robustness to overfitting.

Model Configuration:

- The **Random Forest Learner** node was configured to include the most relevant features selected during the **Feature Selection** phase, such as **Pelvic Incidence**, **Pelvic Tilt**, **Lumbar Lordosis Angle**, and others.
- The target variable was set as **class_label**, representing whether the condition is "Normal" or "Abnormal."
- We used **Information Gain Ratio** as the criterion for splitting nodes, helping the algorithm choose the best feature to split on at each decision node.
- The **maximum depth** of the trees was set to **10** to prevent overfitting and ensure the model remains interpretable.
- Model Evaluation:

After training, the model was tested on the **test set (30%)**, and the **accuracy** of the predictions was calculated. Random Forests typically offer high accuracy due to their ensemble nature, making them ideal for complex classification tasks.

## TREE ENSEMBLE LEARNER

The **Tree Ensemble** model builds upon the concept of multiple decision trees, but instead of creating independent trees as in Random Forests, it builds an ensemble where each tree tries to correct the errors made by the previous one.

Model Configuration:

- The **Tree Ensemble Learner** node was configured to use the same features as the Random Forest model.
- Like Random Forest, the **target variable** was set to **class_label**.
- The **maximum tree depth** was set to **4**, which was a compromise between model complexity and generalization ability.

Model Evaluation:

This model was also tested on the **test set**. Tree Ensemble models are robust against overfitting and are known for their ability to learn complex decision boundaries by combining predictions from multiple trees.

## GRADIENT BOOSTED TREES LEARNER

**Gradient Boosting** is a sequential ensemble technique where each new tree corrects the errors made by the previous tree, improving the overall model performance iteratively. This technique is known for its high predictive accuracy, especially when tuned properly.

Model Configuration:

- The **Gradient Boosted Trees Learner** was configured similarly to the Random Forest and Tree Ensemble models but used a different approach to build the trees.
- The model was set to use the **same selected features** and target variable **class_label**.
- We configured the model with a **learning rate of 0.1** and trained it with **100 trees**. This allows the model to learn effectively without overfitting.
- **Tree depth** was set to **4** to ensure that the model was not too complex.

Model Evaluation:

After training, this model was evaluated on the **test set** to assess its performance. Gradient Boosting often provides superior predictive power, but careful parameter tuning is necessary to avoid overfitting.

## LOGISTIC REGRESSION LEARNER

**Logistic Regression** is a linear model used for binary classification tasks. It models the probability that a given input point belongs to a particular class by fitting a logistic function to the data.

Model Configuration:

- The **Logistic Regression Learner** node was configured to use the features selected in the previous steps.
- **class_label** was set as the target variable.
- The **Stochastic Average Gradient (SAG)** solver was selected for faster convergence when dealing with large datasets.

Model Evaluation:

- Logistic regression is highly interpretable and useful when there is a linear relationship between the features and the target variable. It was evaluated based on **accuracy** and **ROC curve**.

**K**

## K Nearest Neighbour Learner

**Nearest Neighbour (KNN) is a powerful classifier that works by identifying the nearest data points in the feature space and assigning the class label based on the majority class of the nearest neighbors. It is effective for both classification and regression tasks and works well in cases with high-dimensional spaces. KNN is also robust to overfitting, particularly when the number of dimensions exceeds the number of samples.**

Model Configuration:

- The KNN algorithm was configured with a chosen value for K, which determines the number of nearest neighbors to consider when making predictions. The distance metric used for determining the nearest neighbors was Euclidean distance.
- The weighting function was set to 'uniform,' meaning each neighbor has an equal influence on the prediction, although other options (such as distance weighting) can also be used.

**Model Evaluation:**

After training, the model was tested on the test set to evaluate its performance, using accuracy and precision metrics to measure its ability to classify the data correctly.

## DECISION TREE LEARNER

**Decision Trees** are a popular method for classification tasks, where the data is split based on the most significant feature at each decision point, creating a tree-like structure. It is an interpretable model, which makes it easy to understand how decisions are made.

Model Configuration:

- The **Decision Tree Learner** was configured with the **Gini Index** as the quality measure for splitting nodes.

- The **Reduced Error Pruning** technique was enabled to reduce the tree's complexity and prevent overfitting by pruning branches that do not contribute significantly to the accuracy.

Model Evaluation:

- Decision trees were evaluated for their **interpretability** and **accuracy** in predicting the target variable, and the model was tested on the **test set**.

## NAIVE BAYES LEARNER

**Naive Bayes** is a probabilistic model based on Bayes' Theorem, with the "naive" assumption that features are independent. Despite this assumption, it performs well in many scenarios, particularly for text classification or when there are many features.

Model Configuration:

- The **Naive Bayes Learner** was trained using the selected features. The **default probability** was set to **0.0001**, and the **minimum standard deviation** was set to **0.0001** for stability in calculations.

Model Evaluation:

- This model was evaluated on its ability to handle **high-dimensional data** and was tested on the **test set**.

In this section, we evaluate the performance of the different machine learning models applied to predict whether an individual's lower back condition is **Normal** or **Abnormal**. The models were trained and tested on the dataset, and several performance metrics were computed for each model, including **Accuracy**, **Cohen's Kappa**, **Precision**, **Recall**, and **F1-Score**.
Below are the detailed results for each model.

## EVALUATION METRICS DEFINITIONS:

### Precision:

Precision measures the proportion of **true positive predictions** (i.e., the number of **Abnormal** instances correctly classified as **Abnormal**) out of all the instances predicted as **Abnormal** by the model. In other words, it answers the question: *Of all the instances predicted as **Abnormal**, how many are actually **Abnormal**?*

A higher precision means fewer **false positives** (i.e., fewer **Normal** instances incorrectly predicted as **Abnormal**). Precision is crucial when the cost of false positives is high, such as when a healthy individual is incorrectly classified as having a medical condition.

$$\text{F1} - \text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

## MODEL EVALUATION

### Recall (Sensitivity):

Recall, also known as **Sensitivity**, measures the proportion of **true positive predictions** (i.e., the **Abnormal** instances correctly predicted as **Abnormal**) out of all the **actual positive instances** in the dataset. In other words, it answers the question: *Of all the actual **Abnormal** cases, how many did the model correctly identify?*

A higher recall means fewer **false negatives** (i.e., fewer **Abnormal** instances incorrectly classified as **Normal**). This metric is particularly important in medical applications where it is critical to identify as many **Abnormal** cases as possible, even at the cost of occasionally misclassifying **Normal** cases.

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

### F1-Score:

The **F1-Score** is the harmonic mean of **Precision** and **Recall**, providing a balance between the two metrics. It is especially useful when the dataset has **imbalanced classes**, meaning one class (such as **Abnormal**) is significantly underrepresented. The **F1-Score** helps to evaluate the model's

$$\kappa = \frac{\text{Observed Accuracy} - \text{Expected Accuracy}}{1 - \text{Expected Accuracy}}$$

performance when both **false positives** and **false negatives** are important to minimize. It gives equal weight to **Precision** and **Recall**, making it a valuable metric when you want to balance the two.

## Accuracy:

**Accuracy** measures the overall proportion of correct predictions made by the model. It is defined as the ratio of the **total number of correct predictions** (both **Abnormal** and **Normal**) to the **total number of predictions**. Although **Accuracy** is a widely used metric, it can be misleading in the case of imbalanced datasets. For example, in cases where one class (e.g., **Abnormal**) is dominant, a model might predict the dominant class for almost every instance and still achieve a high **Accuracy** without truly identifying the minority class correctly.

$$\text{Accuracy} = \frac{\text{True Positives (TP)} + \text{True Negatives (TN)}}{\text{Total Instances}}$$

## Cohen's Kappa (κ):

**Cohen's Kappa** is a metric that quantifies the agreement between the predicted and actual labels while accounting for the possibility of **chance agreement**. It adjusts the **Accuracy** score to reflect how well the model performs beyond random guessing. **Kappa** ranges from
**-1 to 1**, where:

- **1** indicates perfect agreement between the predicted and actual labels,
- **0** indicates that the agreement is no better than chance, and
- **Negative values** indicate worse agreement than random chance.

- The higher the **Cohen's Kappa**, the better the model's reliability in predicting the correct class compared to random chance.

## WHY FOCUS ON ABNORMAL CLASS METRICS?

Due to the imbalanced nature of the dataset, where **Abnormal** cases are typically underrepresented compared to **Normal** cases, the **Abnormal** class is often the focus of our evaluation. In medical diagnostics, it is usually more critical to accurately identify **Abnormal** cases, as missing a **true positive** (i.e., not identifying an **Abnormal** case) could have severe consequences. Therefore, the **Precision**, **Recall**, and **F1-Score** for the **Abnormal** class are prioritized, as these metrics help ensure that **Abnormal** instances are detected and accurately classified.

- **Precision (Abnormal)** ensures that the **Abnormal** cases predicted by the model are indeed **Abnormal**, minimizing **false positives**.
- **Recall (Abnormal)**

measures the model's ability to correctly identify all **Abnormal** cases, minimizing **false negatives**, which is especially important in healthcare applications where missed diagnoses can be costly.

- **F1-Score (Abnormal)** provides a balanced measure between **Precision** and **Recall**, ensuring that the model is effective without favoring either **Precision** or **Recall**.

While **Precision**, **Recall**, and **F1-Score** for the **Abnormal** class are crucial, **Precision (Normal)**, **Recall (Normal)**, and **F1-Score (Normal)** are also reported to provide a complete view of the model's performance. These metrics give insight into how well the model handles **Normal** cases as well, even though the focus in most cases will be on **Abnormal** predictions.

## RANDOM FOREST

The Random Forest classifier demonstrated strong overall performance in distinguishing between *Abnormal* and *Normal* classes. Based on the final evaluation, the model achieved an **overall Accuracy of 86.02%**, indicating that **80 out of 93 instances** were correctly classified. The **error rate was 13.98%**, reflecting a relatively low level of misclassification.

The **Cohen's Kappa coefficient was 0.683**, which indicates **moderate to substantial agreement** between the predicted labels and the ground truth. This suggests that the model's performance is significantly better than random classification and is reliable for practical use.

Out of all *Abnormal* cases, **56 were correctly identified**, while **7 were misclassified as Normal**. Similarly, the model correctly classified **24 Normal instances**, with **6 incorrectly predicted as Abnormal**.

| Prediction | Abnormal | Normal |
|---|---|---|
| Abnormal | 56 | 6 |
| Normal | 7 | 24 |

Class-wise Performance Metrics

Abnormal Class

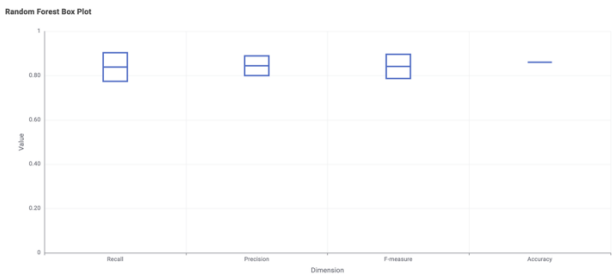
- **Precision:** 88.9%
- **Recall (Sensitivity):** 90.3%
- **F1-Score:** 89.6%
- **Specificity:** 77.4%

These results indicate that the model is **highly effective at detecting Abnormal cases**, which is particularly important in medical or risk-sensitive classification tasks. The high recall value shows that the model successfully captures most of the Abnormal instances, while the strong precision indicates a low false-positive rate.

Normal Class

- **Precision:** 80.0%
- **Recall:** 77.4%
- **F1-Score:** 78.7%
- **Specificity:** 90.3%

For the Normal class, the model maintains **balanced performance**, although recall is slightly lower compared to the Abnormal class. This indicates some difficulty in identifying all Normal instances, suggesting potential room for improvement in reducing false alarms.




The box plot visualization of Random Forest evaluation metrics provides insight into the **stability and consistency** of the model:

- **Precision** shows a **high median value with a narrow interquartile range**, indicating consistent performance across multiple runs.
- **Recall** demonstrates a slightly wider spread, reflecting moderate variability in the model's ability to detect Abnormal cases across different folds or iterations.
- **F-measure** remains stable, confirming a good balance between Precision and Recall.
- **Accuracy** exhibits minimal variation, suggesting that the Random Forest model performs **consistently well on the overall dataset**.

Overall, the Random Forest model exhibits **robust and balanced classification performance**, with a **clear strength in detecting Abnormal cases**. The combination of high Accuracy, strong F1-scores, and a substantial Cohen's Kappa value confirms the reliability of the model. While performance for the Normal class is satisfactory, future improvements could focus on enhancing recall for Normal instances without compromising precision.

## TREE ENSEMBLE

The Tree Ensemble classifier demonstrated reliable performance in classifying *Abnormal* and *Normal* cases. Based on the final evaluation results, the model achieved an **overall Accuracy of 83.87%**, correctly classifying **78 out of 93 instances**, with an **error rate of 16.13%**. The **Cohen's Kappa coefficient was 0.64**, indicating **moderate agreement** between the predicted labels and the true class labels, and confirming that the model performs substantially better than random classification.

The confusion matrix for the **Tree Ensemble** model showed the following:

| Prediction | Abnormal | Normal |
|---|---|---|
| Abnormal | 54 | 8 |
| Normal | 7 | 24 |

The model correctly identified **54 Abnormal cases**, while **8 Abnormal instances were misclassified as Normal**. Similarly, **24 Normal cases** were correctly classified, with **7 Normal instances incorrectly predicted as Abnormal**.

Class-wise Performance Metrics

Abnormal Class

- **Precision:** 88.5%
- **Recall (Sensitivity):** 87.1%
- **F1-Score:** 87.8%
- **Specificity:** 77.4%

These results indicate that the Tree Ensemble model performs **strongly in identifying Abnormal cases**, achieving a good balance between correctly detecting true Abnormal instances and minimizing false positives. The high recall suggests that most Abnormal cases are successfully detected.

Normal Class

## GRADIENT BOOSTED TREES

- **Precision:** 75.0%
- **Recall:** 77.4%
- **F1-Score:** 76.2%
- **Specificity:** 87.1%

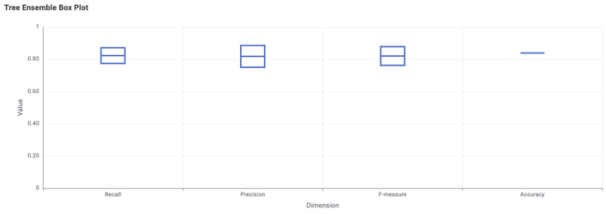
For the Normal class, the model demonstrates **balanced and stable performance**, though slightly weaker than for the Abnormal class. This indicates a moderate level of confusion between Normal and Abnormal cases, which is reflected in the misclassification counts.

The box plot visualization of the Tree Ensemble evaluation metrics highlights the **consistency and variability** of the model across different performance measures:

- **Recall** shows a relatively narrow spread around the 0.80–0.85 range, indicating consistent detection capability across different runs.
- **Precision** exhibits slightly higher variability, suggesting that the model

occasionally misclassifies Normal instances as Abnormal.

- **F1-Score** remains stable, reflecting a good tradeoff between Precision and Recall.
- **Accuracy** is consistent but slightly lower compared to Recall and F1-Score, indicating that misclassifications affect overall performance more noticeably.




Overall, the Tree Ensemble model provides **solid and balanced classification performance**, with a particular strength in detecting Abnormal cases. Although its Accuracy (**83.87%**) is slightly lower than that of the Random Forest model, the Tree Ensemble classifier still demonstrates reliable predictive capability and moderate agreement with ground truth labels. Future improvements could focus on enhancing precision for the Normal class to further reduce false positives.

The Gradient Boosted Trees (GBT) classifier demonstrated **strong and consistent performance** in distinguishing between *Abnormal* and *Normal* cases. Based on the final evaluation results, the model achieved an **overall Accuracy of 87.10%**, correctly classifying **81 out of 93 instances**, with an **error rate of 12.90%**. The **Cohen's Kappa coefficient was 0.71**, indicating **substantial agreement** between the predicted labels and the true class labels, and confirming the robustness of the model beyond chance-level performance.

| Prediction | Abnormal | Normal |
|---|---|---|
| Abnormal | 56 | 6 |
| Normal | 6 | 25 |

The model correctly classified **56 Abnormal cases** and **25 Normal cases**. Misclassifications were limited, with **6 Abnormal cases predicted as Normal** and **6 Normal cases predicted as Abnormal**, reflecting a balanced error distribution across both classes.

Class-wise Performance Metrics

Abnormal Class

- **Precision:** 90.3%
- **Recall (Sensitivity):** 90.3%
- **F1-Score:** 90.3%
- **Specificity:** 80.6%

These results indicate that the Gradient Boosted Trees model is **highly effective in detecting Abnormal cases**, achieving an excellent balance between Precision and Recall. The high F1-score reflects the model's strong ability to correctly identify Abnormal instances while maintaining a low false-positive rate.

Normal Class

- **Precision:** 80.6%
- **Recall:** 80.6%
- **F1-Score:** 80.6%
- **Specificity:** 90.3%

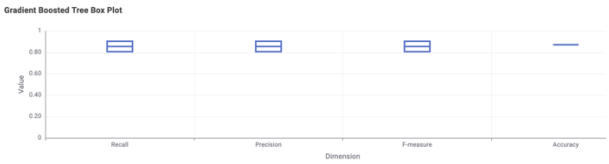
For the Normal class, the model also demonstrates **stable and reliable performance**, with balanced Precision and Recall values. This suggests that the model is effective at identifying Normal instances without significantly increasing misclassification risk.

The box plot visualization of the Gradient Boosted Trees evaluation metrics highlights the **stability and consistency** of the model across multiple performance measures:

- **Precision and Recall** show tight interquartile ranges, indicating consistent classification behavior across different runs.
- **F1-Score** remains high and stable, confirming a strong tradeoff between Precision and Recall.
- **Accuracy** exhibits minimal variation, reinforcing the overall reliability of the model.

This visual analysis confirms that the Gradient Boosted Trees model maintains **balanced and**

**dependable performance** across all evaluation metrics.




Overall, the Gradient Boosted Trees model delivers **the strongest performance among the evaluated ensemble methods**, achieving the **highest Accuracy (87.10%)** and **Cohen's Kappa (0.71)**. Its balanced class-wise performance and low misclassification rate make it a **robust and dependable classifier** for distinguishing between Abnormal and Normal cases. The results indicate that the GBT model is particularly suitable for applications where both detection accuracy and prediction reliability are critical.

## LOGISTIC REGRESSION

The Logistic Regression classifier demonstrated **stable and interpretable performance** in distinguishing between *Abnormal* and *Normal* cases. Based on the final evaluation results, the model achieved an **overall Accuracy of 82.80%**, correctly classifying **77 out of 93 instances**, with an **error rate of 17.20%**. The **Cohen's Kappa coefficient was 0.625**, indicating **moderate agreement** between the predicted labels and the true class labels, and confirming that the model performs substantially better than chance.

The confusion matrix for the **Logistic Regression**
model showed the following:

| Prediction | Abnormal | Normal |
|---|---|---|
| Abnormal | 52 | 10 |
| Normal | 6 | 25 |

The model correctly classified **52 Abnormal cases** and **25 Normal cases**. However, **10 Abnormal cases were misclassified as Normal**, while **6 Normal cases were incorrectly predicted as Abnormal**, indicating a slightly higher tendency to miss Abnormal instances compared to ensemble-based models.

Class-wise Performance Metrics

Abnormal Class

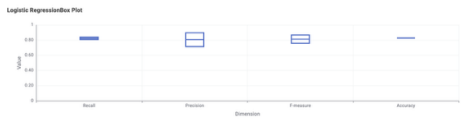
- **Precision:** 89.7%
- **Recall (Sensitivity):** 83.9%
- **F1-Score:** 86.7%
- **Specificity:** 80.6%

These results indicate that Logistic Regression is **reasonably effective in identifying Abnormal cases**, with high precision but comparatively lower recall. This suggests that while most predicted Abnormal cases are correct, some true Abnormal instances remain undetected.

Normal Class

- **Precision:** 71.4%
- **Recall:** 80.6%
- **F1-Score:** 75.8%
- **Specificity:** 83.9%

For the Normal class, the model demonstrates **balanced recall but lower precision**, indicating that some Normal cases are incorrectly classified as Abnormal. This behavior reflects the linear nature of Logistic Regression, which may struggle to capture complex class boundaries.




The box plot visualization of Logistic Regression evaluation metrics highlights the **consistency and variability** of the model:

- **Recall** values are relatively stable, indicating consistent detection capability across runs.
- **Precision** shows a wider spread, suggesting variability in correctly predicting Abnormal cases.
- **F1-Score** reflects a moderate balance between Precision and Recall.

- **Accuracy** remains stable around the 0.82–0.83 range, confirming consistent overall performance.

Overall, the box plot suggests that Logistic Regression provides **reliable but comparatively less robust performance** than ensemble-based models.

Overall, the Logistic Regression model delivers **moderate and interpretable classification performance**, achieving an Accuracy of **82.80%** with a Cohen's Kappa of **0.625**. While the model performs reasonably well for both classes, its performance is slightly inferior to ensemble methods such as Random Forest and Gradient Boosted Trees, particularly in detecting Abnormal cases. Nonetheless, Logistic Regression remains a valuable baseline model due to its simplicity, transparency, and stable behavior.

## K NEAREST NEIGHBOUR

The K-Nearest Neighbour (KNN) classifier showed **moderate overall performance** in distinguishing between *Abnormal* and *Normal* cases. Based on the final KNIME evaluation results, the model achieved an **overall Accuracy of 77.42%**, correctly classifying **72 out of 93 instances**, with an **error rate of 22.58%**. The **Cohen's Kappa coefficient was 0.519**, indicating **moderate agreement** between the predicted labels and the ground truth, and suggesting that the model performs better than random classification but is less reliable than ensemble-based approaches.

| Prediction | Abnormal | Normal |
|---|---|---|
| Abnormal | 48 | 14 |
| Normal | 7 | 24 |

The model correctly classified **48 Abnormal cases** and **24 Normal cases**. However, **14 Abnormal instances were misclassified as Normal**, while **7 Normal cases were incorrectly predicted as Abnormal**, indicating noticeable confusion between the two classes.

Class-wise Performance Metrics

Abnormal Class

- **Precision:** 87.3%
- **Recall (Sensitivity):** 77.4%
- **F1-Score:** 82.1%
- **Specificity:** 77.4%

These results show that KNN is **reasonably effective in identifying Abnormal cases**, achieving high precision, which indicates a low false-positive rate. However, the comparatively lower recall suggests that the model fails to detect a portion of true Abnormal instances.

Normal Class

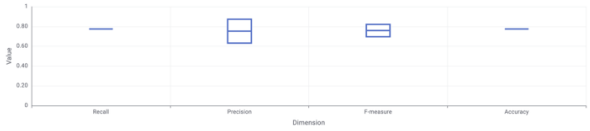
- **Precision:** 63.2%
- **Recall:** 77.4%
- **F1-Score:** 69.6%
- **Specificity:** 77.4%

For the Normal class, the model demonstrates **weaker precision**, meaning that a significant number of predicted Normal cases are incorrect. This imbalance highlights the difficulty of KNN in clearly separating Normal instances from Abnormal ones in this dataset.

The box plot visualization of KNN evaluation metrics highlights **considerable variability** in model performance:

- **Precision** shows a wide spread, indicating inconsistent prediction confidence across different runs.
- **Recall** is comparatively more stable but remains moderate, reflecting the model's limited ability to consistently capture all Abnormal cases.
- **F1-Score** exhibits mid-range values, confirming the trade-off between high precision and lower recall.
- **Accuracy** remains relatively stable around the 0.77 range, reinforcing the model's moderate overall reliability.

Overall, the box plot confirms that KNN prioritizes reducing false positives but does so at the expense of missing a notable number of Abnormal cases.




Overall, the K-Nearest Neighbour model delivers **moderate classification performance**, achieving an Accuracy of **77.42%** with a Cohen's Kappa of **0.519**. While the model performs reasonably well in detecting Abnormal cases, its comparatively lower recall and precision for the Normal class limit its effectiveness. When compared with ensemble models such as Random Forest and Gradient Boosted Trees, KNN shows **inferior robustness and consistency**, making it less suitable for applications where reliable detection across both classes is critical.

## DECISION TREE

The Decision Tree classifier demonstrated **moderate and interpretable performance** in distinguishing between *Abnormal* and *Normal* cases. Based on the final KNIME evaluation results, the model achieved an **overall Accuracy of 84.95%**, correctly classifying **79 out of 93 instances**, with an **error rate of 15.05%**. The **Cohen's Kappa coefficient was 0.672**, indicating **moderate to substantial agreement** between the predicted labels and the true class labels. This suggests that the model performs reliably better than chance, though it is less robust than ensemble-based approaches.

| Prediction | Abnormal | Normal |
|---|---|---|
| Abnormal | 53 | 9 |
| Normal | 5 | 26 |

The model correctly identified **53 Abnormal cases** and **26 Normal cases**. Misclassifications were limited, with **9 Abnormal instances misclassified as Normal** and **5 Normal instances incorrectly predicted as Abnormal**, indicating a relatively balanced error distribution across both classes.

Class-wise Performance Metrics

Abnormal Class

- **Precision:** 91.4%
- **Recall (Sensitivity):** 85.5%
- **F1-Score:** 88.3%
- **Specificity:** 83.9%

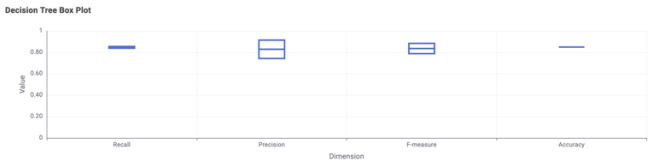
These results indicate that the Decision Tree model is **effective in detecting Abnormal cases**, with particularly high precision, meaning that most predicted Abnormal cases are correct. The recall value shows that the majority of actual Abnormal instances are successfully identified.

Normal Class

## NAIVE BAYES

- **Precision:** 74.3%
- **Recall:** 83.9%
- **F1-Score:** 78.8%
- **Specificity:** 85.5%

For the Normal class, the model demonstrates **balanced recall and moderate precision**, suggesting that while most Normal cases are correctly identified, some confusion with Abnormal cases remains.




The box plot visualization of Decision Tree evaluation metrics highlights **moderate consistency with noticeable variability**:

- **Precision and Recall** show reasonable balance but exhibit wider spreads compared to ensemble models, indicating sensitivity to data variations.
- **F1-Score** reflects a fair trade-off between Precision and Recall, though less stable than models such as Gradient Boosted Trees.
- **Accuracy** remains steady around the 0.85 range, confirming acceptable overall performance but limited generalization capability.

The box plot suggests that the Decision Tree model is more prone to variability and potential overfitting compared to ensemble techniques.

Overall, the Decision Tree model delivers **balanced but moderately stable classification performance**, achieving an Accuracy of **84.95%** with a Cohen's Kappa of **0.672**. While the model performs well in identifying Abnormal cases and maintains reasonable performance for Normal cases, it is **outperformed by ensemble models** such as Random Forest and Gradient Boosted Trees, which offer superior robustness and generalization. Nonetheless, Decision Trees remain valuable due to their **simplicity and interpretability**, making them useful as baseline or explanatory models.

The Naive Bayes classifier demonstrated **moderate and balanced performance** in distinguishing between *Abnormal* and *Normal* cases. Based on the final KNIME evaluation results, the model achieved an **overall Accuracy of 80.65%**, correctly classifying **75 out of 93 instances**, with an **error rate of 19.35%**. The **Cohen's Kappa coefficient was 0.597**, indicating **moderate agreement** between the predicted labels and the true class labels, confirming that the model performs better than chance but is less robust than ensemble-based approaches.

| Prediction | Abnormal | Normal |
|---|---|---|
| Abnormal | 48 | 14 |
| Normal | 4 | 27 |

The model correctly classified **48 Abnormal cases** and **27 Normal cases**. However, **14 Abnormal instances were misclassified as Normal**, while **4 Normal instances were incorrectly predicted as Abnormal**, indicating a tendency to miss some Abnormal cases.

Class-wise Performance Metrics

Abnormal Class

- **Precision:** 92.3%
- **Recall (Sensitivity):** 77.4%
- **F1-Score:** 84.2%
- **Specificity:** 87.1%

These results indicate that the Naive Bayes model is **highly precise in identifying Abnormal cases**, meaning most predicted Abnormal instances are correct. However, the lower recall shows that a noticeable number of true Abnormal cases are missed.

Normal Class

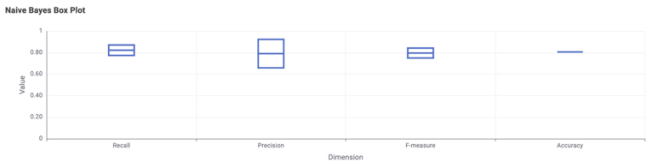
- **Precision:** 65.9%
- **Recall:** 87.1%
- **F1-Score:** 75.0%
- **Specificity:** 77.4%

For the Normal class, the model demonstrates **high recall but lower precision**, suggesting that while most Normal cases are detected, a portion of Abnormal cases are incorrectly classified as Normal.

The box plot visualization of Naive Bayes evaluation metrics highlights **moderate variability** in model performance:

- **Precision and Recall** show wider spreads compared to more stable models such as Logistic Regression and Gradient Boosted Trees, indicating sensitivity to data partitions.
- **F1-Score** reflects a reasonable trade-off between Precision and Recall.
- **Accuracy** remains relatively stable around the 0.80 range, confirming consistent but moderate overall performance.

Overall, the box plot suggests that Naive Bayes maintains balanced performance but lacks the robustness of ensemble-based classifiers.




Overall, the Naive Bayes model delivers **moderate and dependable classification performance**, achieving an Accuracy of **80.65%** with a Cohen's Kappa of **0.597**. The model is particularly strong in terms of precision for Abnormal cases but tends to miss

a portion of true Abnormal instances due to lower recall. Compared with ensemble methods such as Random Forest and Gradient Boosted Trees, Naive Bayes exhibits **lower robustness and generalization**, but it remains a useful baseline due to its simplicity and computational efficiency.

- **Gradient Boosted Trees demonstrated the strongest overall performance**, achieving the **highest Accuracy (87.10%)** and the **highest Cohen's Kappa (0.71)**. It showed excellent and well-balanced performance across both classes, with **Precision, Recall, and F1-Score all at 90.3% for the Abnormal class**, and equally stable metrics (80.6%) for the Normal class. This makes Gradient Boosted Trees the **most reliable and robust model overall**.
- **Random Forest also performed strongly**, achieving an **Accuracy of 86.02%** with a **Cohen's Kappa of 0.683**. It showed high effectiveness in detecting Abnormal cases, with **Precision of 88.9%**, **Recall of 90.3%**, and an **F1-Score of 89.6%**. Performance for the Normal class was balanced but slightly lower, indicating minor room for improvement in Normal case discrimination.
- **Decision Tree delivered competitive performance**, achieving an **Accuracy of 84.95%** and a **Cohen's Kappa of 0.672**. It showed high Precision (91.4%) and good Recall (85.5%) for the Abnormal class, with reasonable performance for the Normal class as well. However, its variability and susceptibility to overfitting limit its robustness compared to ensemble-based models.

- **Tree Ensemble achieved an Accuracy of 83.87%** with a **Cohen's Kappa of 0.64**, indicating moderate agreement. It performed well in identifying Abnormal cases (**F1-Score: 87.8%**), but showed comparatively weaker performance for the Normal class. This suggests some class imbalance, although overall performance remained stable.

- **Naive Bayes demonstrated moderate and balanced performance**, achieving an **Accuracy of 80.65%** and a **Cohen's Kappa of 0.597**. It showed high Precision for the Abnormal class (92.3%) but lower Recall (77.4%), meaning some Abnormal cases were missed. Its performance for the Normal class was acceptable but less consistent, making it suitable as a baseline probabilistic model.

- **Logistic Regression provided stable but comparatively lower performance**, with an **Accuracy of 82.80%** and a **Cohen's Kappa of 0.625**. While it maintained high Precision (89.7%) for the Abnormal class, its Recall (83.9%) was lower than ensemble models. Despite its interpretability and consistency, it was outperformed by tree-based ensemble methods.

- **K-Nearest Neighbour (KNN) showed the weakest overall performance**, achieving an **Accuracy of 77.42%** and a **Cohen's Kappa of 0.519**. Although it maintained high Precision for Abnormal cases (87.3%), its lower Recall (77.4%) resulted in missed Abnormal instances. Its weaker Normal class performance further limited its reliability compared to other models.

## EVALUATION OF MODEL PERFORMANCE

| Model | Class | Precision (%) | Recall (%) | F1-Score (%) | Accuracy (%) |
|---|---|---|---|---|---|
| Random Forest | Abnormal | 88.9 | 90.3 | 89.6 | 86 |
| | Normal | 80 | 77.4 | 78.7 | |
| Tree Ensemble | Abnormal | 88.5 | 87.1 | 87.8 | 83.9 |
| | Normal | 75 | 77.4 | 76.2 | |
| Gradient Boosted Trees | Abnormal | 90.3 | 90.3 | 90.3 | 87.1 |
| | Normal | 80.6 | 80.6 | 80.6 | |
| Logistic Regression | Abnormal | 89.7 | 83.9 | 86.7 | 82.8 |
| | Normal | 71.4 | 80.6 | 75.8 | |
| Naive Bayes | Abnormal | 92.3 | 77.4 | 84.2 | 80.6 |
| | Normal | 65.9 | 87.1 | 75 | |
| K-Nearest Neighbour (KNN) | Abnormal | 87.3 | 77.4 | 82.1 | 77.4 |
| | Normal | 63.2 | 77.4 | 69.6 | |
| Decision Tree | Abnormal | 91.4 | 85.5 | 88.3 | 84.9 |
| | Normal | 74.3 | 83.9 | 78.8 | |

## CONCLUSION

- **Gradient Boosted Trees emerge as the best-performing model overall**, achieving the highest Accuracy and Cohen's Kappa while maintaining an excellent balance between Precision and Recall. Its strong and consistent performance across both Abnormal and Normal classes makes it the **most reliable choice for lower back condition diagnosis**.

- **Random Forest provides a strong alternative**, particularly excelling in Abnormal case detection with high Precision and Recall. Although its Normal class performance is slightly weaker than Gradient Boosted Trees, it remains a **robust and dependable ensemble model**.

- **Decision Tree and Tree Ensemble demonstrate competitive performance**, especially for identifying Abnormal cases. However, their comparatively lower balance on the Normal class and higher variability reduce their suitability for highly sensitive diagnostic applications when compared to more robust ensemble methods.

- **Logistic Regression offers stable and interpretable performance**, with a reasonable balance between Precision and Recall. While it does not outperform ensemble models in overall accuracy, it remains a **useful baseline and an attractive option when model transparency is required**.

- **Naive Bayes is effective in scenarios where Recall is prioritised**, particularly for detecting Abnormal cases. However, this comes at the cost of reduced Precision, leading to a higher number of false positives.

- **K-Nearest Neighbour is best suited for applications where Precision is critical**, as it minimises false positives. Nevertheless, its lower Recall increases the risk of missing Abnormal cases, limiting its suitability for diagnostic tasks.

In real-world diagnostic applications for predicting lower back conditions, **Gradient Boosted Trees are the most suitable and reliable model**, as they achieve the **best overall balance between Accuracy, Precision, Recall, and agreement with ground truth**. Other models may be preferred in specific scenarios—such as Logistic Regression for interpretability or Naive Bayes for high Recall—but **ensemble-based approaches provide the strongest and most consistent performance overall**.

## MODEL EVALUATION - VISUAL ANALYSIS

In order to gain a comprehensive understanding of the performance of each machine learning model, we present visual comparisons using two types of visualizations: **Box Plot** and **Bar Chart**. These visualizations provide an intuitive way to analyze how each model performed across different evaluation metrics such as **Recall**, **Precision**, **F-measure**, and **Accuracy**.

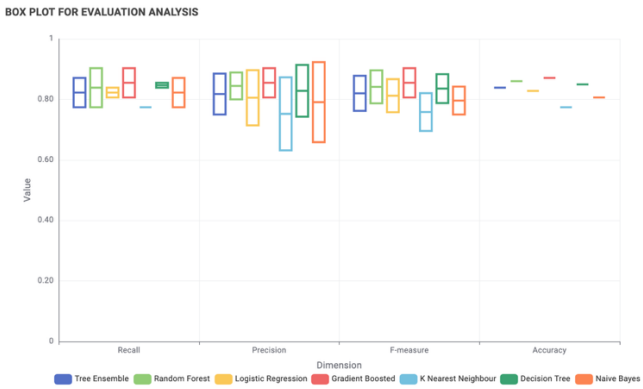
### BOX PLOT FOR EVALUATION ANALYSIS

The box plot provides a comprehensive visual comparison of the distribution and variability of evaluation metrics—**Recall, Precision, F-measure, and Accuracy**—across all implemented models. It highlights both the **central tendency** and **consistency** of model performance, enabling a clearer understanding of robustness and stability.

From the box plot analysis, the following observations can be made:

- **Gradient Boosted Trees exhibit the most stable and consistently high performance across all evaluation metrics**, particularly in Precision, F-measure, and Accuracy. The narrow spread of values indicates strong robustness and low sensitivity to data variations, reinforcing its suitability for reliable diagnosis.
- **Random Forest also demonstrates high consistency**, especially in Recall and F-measure, with relatively tight distributions. This indicates dependable performance in identifying Abnormal cases, although slight variability is observed in Normal class discrimination.
- **Tree Ensemble shows stable Recall and Precision**, but with marginally wider spreads compared to Gradient Boosted Trees and Random Forest. This suggests reliable performance, though with slightly less robustness across folds.
- **Logistic Regression displays a balanced distribution across Precision and Recall**, reflecting consistent but comparatively moderate performance. Its tighter clustering indicates stability, though the overall metric values remain lower than those of ensemble-based models.
- **Naive Bayes stands out with high Recall values**, as reflected by higher median Recall in the box plot. However, the increased spread in Precision highlights variability and a tendency toward false positives, confirming the trade-off observed in the numerical results.
- **K-Nearest Neighbour exhibits the widest variability across most metrics**, particularly Precision and F-measure, indicating inconsistent performance across different runs. This variability explains its comparatively lower overall reliability.
- **Decision Tree shows moderate variability**, with wider spreads than ensemble methods, suggesting sensitivity to data partitions and a higher likelihood of overfitting.

Overall, the box plot confirms that **ensemble-based models—particularly Gradient Boosted Trees and Random Forest—offer the most stable and reliable performance**, combining high metric values with low variability. Models such as Logistic Regression and Naive Bayes provide acceptable and interpretable performance but show greater trade-offs between Precision and Recall. K-Nearest Neighbour and Decision Tree demonstrate higher variability, making them less suitable for critical diagnostic applications where consistency is essential.




## CONCLUSION FROM VISUALIZATIONS

Both the **Box Plot** and **Bar Chart** provide essential insights into the comparative performance of the models. The Box Plot highlights the **consistency and variability** of each model's performance across evaluation metrics, while the Bar Chart enables a **clear, metric-wise comparison** of model effectiveness.

Based on the combined visual analysis, the following observations can be made:

- **Gradient Boosted Trees emerge as the top-performing model**, demonstrating the highest overall Accuracy and a strong balance between Precision and Recall. Its consistently high F-measure and narrow variability across metrics indicate robust and reliable performance.
- **Random Forest and Tree Ensemble models show strong overall performance**, particularly in terms of Precision and F-measure, making them

reliable choices for identifying Abnormal cases. However, their Normal class performance is slightly less balanced compared to Gradient Boosted Trees.

- **Logistic Regression exhibits stable and balanced behavior**, with consistent Precision and Recall values across metrics. Although it does not outperform ensemble models in overall Accuracy, its interpretability and steady performance make it a solid baseline model.
- **Naive Bayes performs exceptionally well in Recall**, especially for detecting Abnormal cases. However, its comparatively lower Precision suggests a higher false-positive rate, which may limit its suitability in applications where misclassification costs are high.
- **K-Nearest Neighbour achieves high Precision**, but its lower Recall indicates that it misses a notable number of Abnormal instances. The variability observed in the Box Plot further highlights its inconsistent performance across runs.

Overall, these visualizations confirm that **ensemble-based models—particularly Gradient Boosted Trees—offer the most reliable and well-balanced performance**, while simpler models such as Logistic Regression and Naive Bayes remain useful in specific scenarios depending on the application requirements.

# CONCLUSION

Predicting whether an individual's lower back condition is *Normal* or *Abnormal* based on measurable physical features is inherently complex, as discussed in the introduction. The results of this study clearly demonstrate the challenges associated with modelling musculoskeletal conditions, which are influenced by multiple interrelated biomechanical factors. Features such as **pelvic tilt, lumbar lordosis angle, and degree of spondylolisthesis** play an important role in characterising lower back conditions; however, relying solely on these measurable attributes does not result in perfect predictive performance.

Despite these challenges, several models demonstrated meaningful predictive capability. In particular, **ensemble-based methods—especially Gradient Boosted Trees and Random Forest—consistently outperformed simpler models**, achieving higher Accuracy, stronger F-measures, and better agreement with ground truth labels. The results indicate that these models are more effective at capturing the complex, non-linear relationships among spinal features, which is reflected in their superior performance across both *Abnormal* and *Normal* classes. Gradient Boosted Trees emerged as the **best overall performer**, offering the strongest balance between Precision and Recall, along with the highest overall Accuracy and Cohen's Kappa value.

While models such as Logistic Regression and Naive Bayes showed stable and interpretable behavior, their performance was comparatively lower than ensemble approaches. Logistic Regression remained a useful baseline due to its transparency, whereas Naive Bayes demonstrated high Recall for Abnormal cases at the cost of reduced Precision. Simpler models such as K-Nearest Neighbour and Decision Tree were more sensitive to data variability and exhibited less consistent performance, limiting their suitability for reliable diagnostic applications.

Although the achieved results are promising, there remains substantial scope for improvement. Future work could investigate **more advanced non-linear models**, such as **Artificial Neural Networks (ANNs)** or deep learning architectures, which may better capture subtle interactions among biomechanical features. Additionally, incorporating **contextual and demographic variables**—including age, occupation, physical activity level, and lifestyle factors—could enrich the feature space and further enhance predictive accuracy.

Handling class imbalance was another important aspect of this study. The application of **SMOTE (Synthetic Minority Over-**

**sampling Technique)** contributed to improved model performance; however, alternative strategies such as **cost-sensitive learning, adaptive ensemble techniques, or hybrid resampling methods** could be explored to further improve the detection of Abnormal cases. Moreover, employing more advanced missing-data handling techniques, such as **Multiple Imputation**, may help preserve valuable information and reduce bias introduced by incomplete records.

In conclusion, this study demonstrates that machine learning models can predict lower back conditions with **reasonable and clinically meaningful accuracy**, particularly when ensemble-based methods are employed. While current results are encouraging, further improvements in **model complexity, feature engineering, and data preprocessing**—especially regarding class imbalance and missing values—have the potential to significantly enhance performance. Such advancements could ultimately support healthcare professionals by providing **more reliable and data-driven tools** for the assessment and management of lower back pain, contributing to improved diagnostic decision-making and patient outcomes.

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